Phase 2 - Fundamentals of Natural Language Processing

Natural Language Processing

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Abstract

This report presents the application of Natural Language Processing (NLP) techniques to sentiment analysis and information retrieval through two computational experiments. The first experiment analyzes a mental health-related text dataset containing approximately 53,000 statements classified into seven categories. It combines lexicon-based sentiment analysis using VADER with supervised and semi-supervised learning approaches, including Logistic Regression, Linear Support Vector Classification (SVC), and Complement Naïve Bayes. A label propagation strategy based on model agreement is also evaluated. The second experiment implements and compares information retrieval models using the CISI benchmark collection, comprising 1,460 documents and 112 queries. The evaluation considers TF-IDF, BM25, Latent Semantic Analysis (LSA), word embeddings, and a hybrid ranking approach, examining the effects of text preprocessing, stemming, lemmatization, and document representation. Results show that VADER identifies general sentiment patterns but cannot reliably distinguish sentiment polarity from mental health status. The semi-supervised approach achieves an accuracy of 0.740 and a macro-F1 score of 0.684, although its improvement over training with the labeled subset is not statistically established. In information retrieval, stemming improves mean average precision (MAP), increasing TF-IDF performance from 0.197 to 0.229 and BM25 from 0.181 to 0.225. However, differences among the strongest retrieval systems are generally not statistically significant. Overall, the findings highlight the importance of preprocessing, feature representation, and evaluation design in NLP applications, while emphasizing the limitations of automated text classification and the need for cautious interpretation in sensitive contexts.

Introduction

Natural language processing (NLP) is a subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate using human language. It also enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics, the rule-based modeling of human language together with statistical modeling, machine learning and deep learning. [1]

Among its most relevant applications are sentiment analysis and information retrieval, which support the interpretation

of textual content and the efficient discovery of relevant information. However, each technique has a different approach to analyze text: one seeks to understand the "how is it said" (the emotion), while the other seeks to find the "what does it says" (the content).

Sentiment analysis (Also known as opinion mining) is an NLP task focused on identifying and classifying subjective information expressed in text, such as opinions, emotions, and attitudes. Depending on the application, sentiment analysis may involve assigning predefined categories to textual statements or discovering underlying patterns without previously labeled examples [2]. In this report we will be exploring both approaches.

In mental health-related contexts, these techniques can be applied to explore linguistic patterns and relationships between textual expressions and mental health categories. However, it is important to note that automated classifications should be interpreted with caution. This is because text alone cannot establish a clinical diagnosis or fully represent an individual's psychological condition, therefore a professional's opinion is still necessary.

Information retrieval (IR), in contrast, focuses on identifying and ranking documents according to their relevance to a user's information needs. Traditional IR systems commonly rely on lexical representations and statistical weighting techniques, such as Term Frequency–Inverse Document Frequency (TF-IDF), combined with similarity measures to compare queries against documents. More advanced approaches incorporate semantic representations and machine learning to improve retrieval effectiveness. The evaluation of IR systems requires appropriate metrics, such as precision, recall, and F1-score, to measure their ability to retrieve relevant documents.

In this case, the CISI collection is a classic benchmark for information retrieval research, comprising 1,460 documents, 112 queries, and relevance assessments linking queries to relevant documents. Its structure makes it suitable for developing and evaluating retrieval systems under controlled conditions.

Unlike sentiment analysis, which focuses on identifying characteristics or categories within individual texts, information retrieval aims to identify documents that best satisfy a query. The CISI collection enables students to investigate how document preprocessing, term weighting, vector representations, and similarity measures influence retrieval performance.

Objectives

General objective

To design, implement, and evaluate Natural Language Processing techniques for sentiment analysis and information retrieval using the mental health corpus and the CISI benchmark collection, applying text preprocessing, feature representation, unsupervised and supervised learning, and quantitative evaluation methods to identify textual patterns, classify statements, and retrieve relevant documents effectively.

Specific objectives

- To apply appropriate text cleaning, normalization, tokenization, and

feature extraction techniques to prepare both datasets for computational analysis.
- To identify potential patterns, similarities, and groupings among mental health-related statements without relying exclusively on predefined labels.
- To train and assess suitable machine learning algorithms using appropriate evaluation metrics
- To represent documents and queries using appropriate textual features and implement a retrieval method that ranks documents according to their estimated relevance to each query.
- To compare the performance of the implemented methods, discuss the influence of preprocessing and representation choices, identify methodological limitations, and formulate recommendations for potential improvements in sentiment analysis and information retrieval.

Methodology

1 Sentiment Analysis on mental's health dataset:

The "Sentiment Analysis for Mental Health" dataset from Kaggle used in this analysis consists of around 53.043 statements tagged with one of the following seven mental health statuses: Normal, Depression, Suicidal, Anxiety, Stress, Bi-Polar, and Personality Disorder.

The data is sourced from diverse platforms including social media posts, Reddit posts, Twitter posts, and more. Each entry is tagged with a specific mental health status, making it an invaluable asset for training machine learning models aimed at understanding and predicting mental health conditions based on textual data. [3]

Initially, the columns of interest are selected: the statements and the associated label in the status column. Subsequently, empty values are removed from these columns to ensure the dataset contains no incomplete records. Regarding the statements, leading and trailing whitespace is removed, and empty and duplicated statements are eliminated.

Following data preprocessing, the number of records was reduced to 51,073; the texts had an average length of approximately 112.7 words, ranging from 1 to 6,300 words. Furthermore, the largest identified category was "normal states" (approximately 31.4%), followed by "depression" (29.5%) and "suicidal" states (20.8%), while the remaining states accounted for a smaller percentage.

For the unsupervised, lexicon-based sentiment analysis we used the VADER (Valence Aware Dictionary and sEntiment Reasoner) method since it is built for English and tuned on social-media text, which matches the Reddit/Twitter-style origin of this corpus. This model is tuned for informal wording, capitalisation, repeated punctuation and emoticons. Also, it is important to note that the model does not need to training data, it's fast, and interpretable.

However, since the model is meant for general purpose, and distress is often expressed without any sentiment-bearing words ("I don't see the point anymore"), through sarcasm, or via clinical vocabulary absent from the lexicon; the results may need additional analysis before jumping into conclusions.

Although every row has a label, for the purposes of an exercise involving semi-supervised models, we will assume that only a small, random subset is labeled. Next, to train and evaluate a model, we will follow these steps:

1. Randomly draw a labeled pool of 3000 documents. To split the data subsets, we use a stratification to ensure that the proportion of each class remains identical in both the training set and the test set.

status	train	gold_test
Normal	678	291
Depression	602	258
Suicidal	422	180
Anxiety	150	64
Bipolar	109	47
Stress	98	42
Personality disorder	41	18

2. Split the pool into two disjoint subsets: a training subset (labels used) and a gold-standard test subset. In this case, labels were used only for evaluation; during propagation they are hidden like the rest of the corpus.
3. We assume that everything outside the pool is the unlabeled corpus. As mentioned before, its original labels are kept aside, and they are used only for extra validation.
4. Finally, as an example, we will propagate the labels with three different methods using different models, then measure their accuracy on the gold test set. The models we are going to use are:

a. SVC and logistic regression on the labeled subset and use it to label the remaining corpus.

Before training the models, we will use the TF-IDF interpretation using unigrams and bigrams fitted on the texts only. Also, the URLs are replaced by a `url` token. As for the stop-words, we will keep them in this way negations and first-person pronouns carry signal here.

Now for the tuning protocol of the models, we will not use the gold test set for tuning. In other hand, the train subset is split 70/30 training and testing respectively.

We pick the feature set (unigrams vs. unigrams+bigrams) and the regularization strength `C` / `alpha` by macro-F1 on the training subset. This decision was made because macro-F1 handles the class imbalance due to the size of the dataset.

b. The second methodology was based on fine-tuning and semi-supervised retraining, employing the Label Spreading technique. Initially, three combinations of hyperparameters were evaluated—specifically regarding the number of nearest neighbors and the constraint factor (alpha), which controls the extent to which an original label can be altered.

Finally, once the optimal hyperparameter combinations for the model were determined, the labels were propagated from the training data subset to the remaining nodes.

c. The final methodology used was focused on obtaining a conservative corpus by agreement. In this case, we assign, for every document, three

predictions and keep the label only if at least two of the three methods agree.

Documents that do not achieve a 2-out-of-3 majority are discarded from the propagated corpus. We present the coverage (proportion retained) and precision for the resulting subset, which reflects the typical trade-off between precision and coverage.

Finally, having three different labeled corpus, we will train three different supervised models which are: Logistic Regression (strong, interpretable linear baseline for sparse text), Linear SVC (the SVC family; linear kernel is the standard choice for TF-IDF) and Complement Naïve Bayes (fast, designed for imbalanced text classes).

Additionally, we will present a Global view — feature importance, because, for the linear models the coefficients are directly interpretable: the largest positive coefficients of a class are the n-grams that push a document towards it. Also, we will present a local view (LIME) on three representative gold-test documents: the most confidently correct prediction for three different classes (preferring Normal, Depression and Suicidal, if present). LIME perturbs the text by removing words and fits a sparse local linear model to show which words push the prediction towards or away from the predicted class.

2 Information retrieval with CISI:

The CISI collection is a classic benchmark for information retrieval research, comprising 1,460 documents, 112 queries, and relevance assessments linking queries to relevant documents. Its structure makes it suitable for developing and evaluating retrieval systems under controlled conditions [4] .

For this dataset, an information retrieval (IR) system will be used that takes a text-based query as input and returns documents ranked according to their relevance.

To achieve this, the documents and queries were converted into numerical representations. Subsequently, a score was calculated for each query-document pair, and the documents were then ordered based on these scores. Finally, the resulting ranking is compared against the relevance judgments from CISI.REL.

Initially, the tagged CISI dataset is loaded. In this process, the .X block (cross-references) is ignored, as it contains no document text that provides information relevant to the task. Each document is indexed using its title and abstract.

Furthermore, a binary relevance matrix (REL) is constructed from the CISI.REL file—covering the queries evaluated against the 1,460 documents—in which a value of 1 is assigned whenever a document is deemed relevant.

Additionally, the consistency across the files is verified to ensure that no identifier present in the REL data is absent from the document collection.

It should be noted that only 76 of the 112 queries have associated relevance judgments; the remaining 36 cannot be evaluated.

For data preprocessing, the text was converted to lowercase; subsequently, hyphens and slashes were separated to facilitate tokenization using the NLTK

library; finally, non-alphanumeric characters were removed.

In this stage, four variants of the text are generated to subsequently compare the performance of each model, with each model having been trained on a different version of the text. The first variant, labeled "raw," consists solely of conversion to lowercase and tokenization. The second variant, "clean," is based on the first but excludes stop words, isolated numbers, and single-character tokens. The third variant, "stem," builds upon the previous one and applies the Porter stemming algorithm. Finally, the fourth variant, "lemma," is based on the second and employs lemmatization via WordNet, guided by part-of-speech (POS) tags.

	vocabulary	tokens / doc	tokens / query
raw	10111.0	128.3	76.5
clean	9543.0	71.0	43.7
stem	5718.0	71.0	43.7
lemma	7649.0	71.0	43.7

```
Example query 1 ->
    raw: ['what', 'problems', 'and', 'concerns', 'are', 'there', 'in', '
  clean: ['problems', 'concerns', 'making', 'descriptive', 'titles', 'di
   stem: ['problem', 'concern', 'make', 'descript', 'titl', 'difficulti'
  lemma: ['problem', 'concern', 'make', 'descriptive', 'title', 'difficu
```

For each evaluated query, we assigned a score to each document; the documents are then ranked in descending order based on these scores. In the event of a tie, the original order of the documents is preserved, ensuring a deterministic result. Assuming binary relevance and a known number of relevant documents, we used the following metrics at each model:

- Precision@k = (number of relevant items among the top k) / k.
- Recall@k = (number of relevant items among the top k) / (total number of relevant items).
- F1@k = the harmonic mean of the two aforementioned metrics.

It is important to note that Recall@k cannot reach a value of 1 when the total number of relevant documents exceeds `k`. Since several queries in the CISI dataset contain more than 100 relevant documents, the absolute values of Precision, Recall, and F1@k are bounded and should be interpreted in relation to one another.

- AP (Average Precision) = the mean of the precision values calculated at the position of each relevant document.
- MAP (Mean Average Precision) = the mean of the AP scores across the entire set of queries; this serves as the primary summary metric.
- nDCG@10, using binary relevance gains and a discount factor of 1/log2(position+1).
- MRR (Mean Reciprocal Rank) = the mean of the reciprocal ranks of the position of the first relevant document.
- Interpolated P–R (Precision–Recall) curve: precision at recall levels ranging from 0.0 to 1.0, averaged across the entire set of queries.

The models trained were TF-IDF + cosine (VSM), BM25 (Okapi), LSA, Word2Vec (trained on CISI corpus), GloVe (pretrained), and a hybrid model using BM25 combined with the best-performing model via Reciprocal Rank Fusion.

Initially, a comparison was conducted to determine the approach to be used for evaluating the remaining models. This included testing TF-IDF and BM25

models—each with four variants—along with an ablation study comparing the use of "summary only" versus "title + summary." Based on these results, it was decided to apply a stemming algorithm during model training; the exception is the GloVe model, for which this process is omitted to ensure that the terms can be matched against the model's pre-trained vocabulary.

Initially, all systems are trained using default parameters, and the results are presented in a table containing all metrics. Additionally, a bootstrap confidence interval for the MAP (based on 5,000 resamples of the queries) is included, along with P–R curves.

Subsequently, to avoid overly optimistic results—which would arise if parameters were tuned using the same 76 queries later employed for reporting results—we apply cross-validation across the queries. Specifically, we use 5-fold cross-validation, repeated 10 times. In each fold, the configuration that achieves the best mean AP on the training queries is selected and then evaluated on the queries held out for testing.

With only 76 queries, small differences could simply be noise. For each pair of systems, the mean paired difference in AP (average precision) is calculated, along with a 95% bootstrap confidence interval (based on 10,000 resamples) and the Wilcoxon signed-rank test.

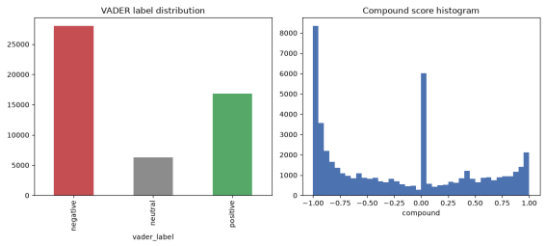
To evaluate the efficiency of the trained models—and given that all the systems employed implement a `score_texts(texts)` function—users can input any query in free-text format. This input undergoes the exact same preprocessing as the documents and is projected into the same space (whether a sparse term space, an LSA space, or a word-vector space) before the scoring process takes place. In this context, an exercise was conducted to observe the performance of each model, specifically with regard to retrieving the top-ranked titles for a given query.

Conclusions and discussions

1 Sentiment Analysis on mental health dataset:

At the initial Unsupervised, lexicon-based sentiment analysis we can see that VADER labels 54.8 % of the 51,073 statements negative, 12.3 % neutral and 32.9 % positive.

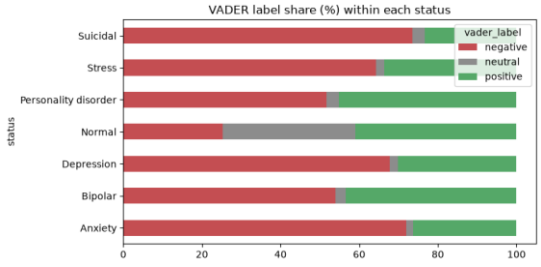
vader_label	n	share_%
negative	28000	54.8
neutral	6282	12.3
positive	16791	32.9



Although the mean composite score decreases monotonically according to the clinical severity of the label (suicidal tendencies: −0.44; anxiety: −0.36; depression: −0.33; stress: −0.26; bipolar disorder: −0.10; personality disorder: −0.06; normal: +0.10), the lexicon manages to capture a real, but imprecise, signal.

vader_label	negative	neutral	positive	mean_compound
status				
Anxiety	71.9	1.8	26.3	-0.363
Bipolar	54.0	2.6	43.4	-0.101
Depression	67.7	2.2	30.1	-0.331
Normal	25.3	33.7	41.0	0.098
Personality disorder	51.6	3.2	45.1	-0.064
Stress	64.2	2.2	33.6	-0.260
Suicidal	73.6	3.1	23.3	-0.442

We can see that it fails often at the individual level: 31.7 % of all non-Normal statements are neutral/positive (about a quarter of Depression, Suicidal and Anxiety posts are labelled positive), while 25.3 % of Normal statements are labelled negative.



Some causes may involve implicit distress without sentiment words, narrative or clinical vocabulary missing from the lexicon, and positive words used in a negative context. We can conclude that the problem may not be by a few missing words, but that polarity is not the same construct as mental-health status.

On the 900-document gold test, logistic regression trained on the propagated corpus reaches accuracy 0.740 and macro-F1 0.684 (Linear SVC 0.736 / 0.680; Complement NB 0.646 / 0.537). Training on only the 2,100 labeled documents gives 0.730 / 0.655, and training on the original labels of the whole corpus gives 0.776 / 0.721.

Gold test: coverage = 97.9% | accuracy on kept = 0.740 | macro-F1 on kept = 0.658
Unlabeled corpus: coverage = 97.9% | accuracy of propagated labels vs hidden originals = 0.724

model	accuracy	precision_macro	recall_macro	f1_macro	f1_micro	f1_weighted	coverage
A1 LogReg	0.730	0.730	0.619	0.655	0.730	0.722	1.000
A2 LinearSVC	0.717	0.722	0.603	0.642	0.717	0.706	1.000
B LabelSpreading	0.631	0.487	0.448	0.453	0.631	0.620	1.000
C Agreement (kept only)	0.740	0.746	0.618	0.658	0.740	0.731	0.979

Propagation therefore recovers roughly a fifth of the accuracy gap and about two fifths of the macro-F1 gap, but the bootstrap 95 % intervals (accuracy [0.711, 0.769], macro-F1 [0.635, 0.727]) overlap with the lower bound, so the gain is suggestive, not statistically established.

Label spreading was the weakest labeler (0.631 / 0.453): the 100-d SVD keeps only ~18 % of the TF-IDF variance, so the k-NN graph is poorly informative, and the 2-of-3 agreement rule (97.9 % coverage) mostly reflects logistic regression and SVC agreeing. A better graph (more components, cosine on TF-IDF directly, or sentence embeddings) is the obvious improvement.

Per-class F1: Normal 0.88, Anxiety 0.77, Depression 0.69, Bipolar 0.67, Personality disorder 0.64 (only 18 test cases), Suicidal 0.63, Stress 0.51.

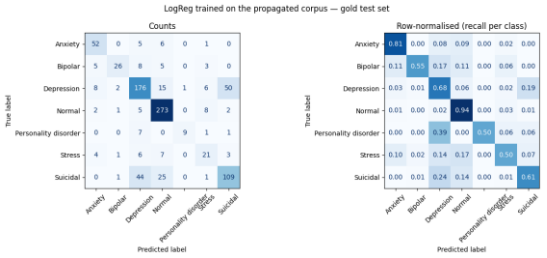
	precision	recall	f1-score	support
Anxiety	0.73	0.77	0.75	64
Bipolar	0.89	0.51	0.65	47
Depression	0.68	0.68	0.68	258
Normal	0.81	0.95	0.87	291
Personality disorder	0.89	0.44	0.59	18
Stress	0.44	0.38	0.41	42
Suicidal	0.67	0.60	0.63	180
accuracy			0.73	900
macro avg	0.73	0.62	0.66	900
weighted avg	0.73	0.73	0.72	900

The two most frequent errors are Depression → Suicidal (50) and Suicidal → Depression (44), about 40 % of all 234 errors: the two classes share vocabulary and differ in degree of intent, which bag-of-words cannot measure. Suicidal → Normal (25 of 180, 14 %) is the clinically worst error type. Stress is confused with almost everything (recall 0.50), and

Personality disorder is mostly absorbed by Depression (7 of 18).

Best model on the propagated corpus (macro-F1): LogReg

	precision	recall	f1-score	support
Anxiety	0.73	0.81	0.77	64
Bipolar	0.84	0.55	0.67	47
Depression	0.70	0.68	0.69	258
Normal	0.82	0.94	0.88	291
Personality disorder	0.90	0.50	0.64	18
Stress	0.51	0.50	0.51	42
Suicidal	0.66	0.61	0.63	180
accuracy			0.74	900
macro avg	0.74	0.66	0.68	900
weighted avg	0.74	0.74	0.73	900



The largest coefficients are condition names, medications and symptom words for each class (anxiety/anxious/restless; manic/mania/lamictal/seroquel; die/suicidal/kill/cannot/myself; avpd/avoidant), while Normal is led by short conversational words.

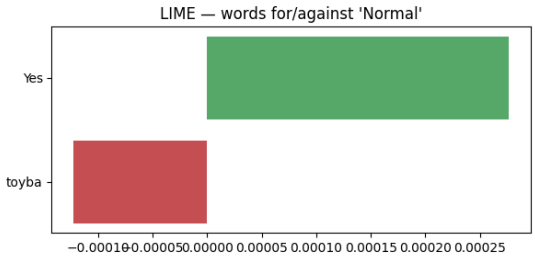
	Anxiety	Bipolar	Depression	Normal	Personality disorder	Stress	Suicidal
0	anxiety	bipolar	depression	you	avpd	stress	am
1	restless	meds	depressed	oh	poll	stressed	die
2	anxious	manic	am	yes	avoidant	he	do not
3	worried	episode	wa	wa	https www	and	not
4	nervous	mania	feel	url	www reddit	https	to die
5	had	ve	friend	twitter	reddit com	our	suicide
6	health anxiety	ve been	just	she	www	him	myself
7	symptoms	seroquel	not	her	people	stress and	cannot
8	blood	lamictal	do	money	poll https	of stress	kill
9	worry	on	life	are you	view poll	com	suicidal
10	health	diagnosed	do not	who	com poll	we	fucking
11	my anxiety	mood	depression and	quot	them	asleep	not want

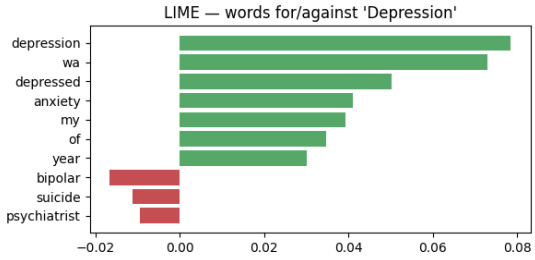
LIME on four gold-test documents confirms the pattern:

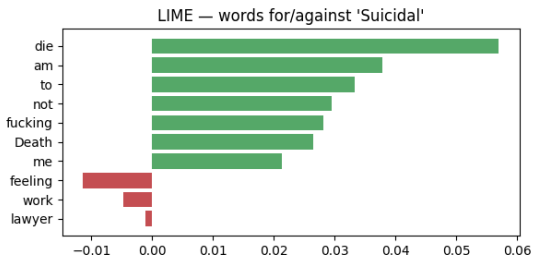
Depression (correct): "depression", "depressed", "anxiety", "my" push towards Depression, although the post lists several diagnoses (comorbidity), so the label is only one of several plausible ones.

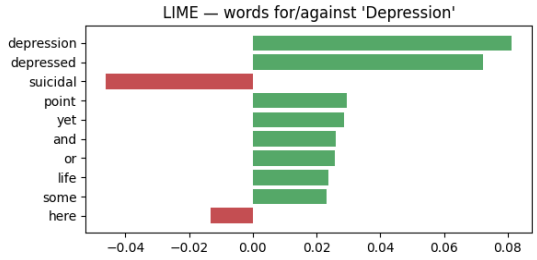
Suicidal (correct): "die", "am", "not", "death" drive the prediction. First-person pronouns and negations/absolutes in suicidal language match findings in the literature on depression and suicidality, although the weights per word are small (≤0.06) because the evidence is spread over many features.

Normal (correct): the example is a two-word message; every LIME weight is ≈0, i.e. the model defaults to the majority/"chat-like" class when there is no clinical vocabulary, which is why Suicidal → Normal errors occur for implicit or indirect expressions.









Error case (true Suicidal, predicted Depression): the author writes that they do not feel depressed but do not want to live. "depression"/"depressed" add +0.08/+0.07 towards Depression, and "suicidal" counts against it: a bag-of-words model has no notion of negation scope, so it follows the surface vocabulary and misses the meaning.

It is important to note that:

The classes seem to be partly separated by their origin (explicit condition names and medications, community-specific terms such as "avpd", conversational chat for Normal), so the 74 % accuracy probably overstates generalization to texts from other platforms.

The texts appear to be pre-processed upstream (tokens like "wa" from "was", "ve" from "I've"): "wa" is a high-weight feature, an artefact, not a signal.

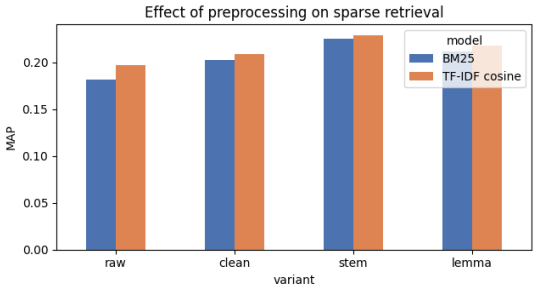
The gold test is small for the rare classes, hence the wide intervals.

These models are educational and must not be used to screen or make decisions about people.

2 Information retrieval with CISI:

Going from raw tokens to stemmed tokens raises TF-IDF MAP from 0.197 to 0.229 (+3.2 points) and BM25 MAP from 0.181 to 0.225 (+4.4), while shrinking the vocabulary from 10,111 to 5,718 terms. Lemmatization (0.218 / 0.212) is better than no normalization but worse than Porter stemming here: stemming conflates more derivational variants (**classification / classify / classified**), which helps recall on short abstracts, and CISI's vocabulary is mostly plain English. Removing stop-words and numbers alone gives +1.2 / +2.1 points.

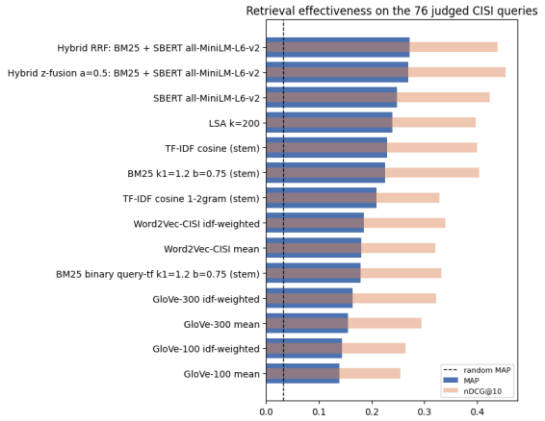
	model	variant	vocab	MAP	nDCG@10	P@10	R@10
0	TF-IDF cosine	raw	10111	0.1972	0.3529	0.2961	0.1245
1	BM25	raw	10111	0.1813	0.3346	0.2895	0.1176
2	TF-IDF cosine	clean	9543	0.2091	0.3663	0.3079	0.1276
3	BM25	clean	9543	0.2023	0.3648	0.3092	0.1363
4	TF-IDF cosine	stem	5718	0.2291	0.4000	0.3539	0.1343
5	BM25	stem	5718	0.2250	0.4046	0.3671	0.1474
6	TF-IDF cosine	lemma	7649	0.2184	0.3759	0.3250	0.1381
7	BM25	lemma	7649	0.2118	0.3881	0.3421	0.1449

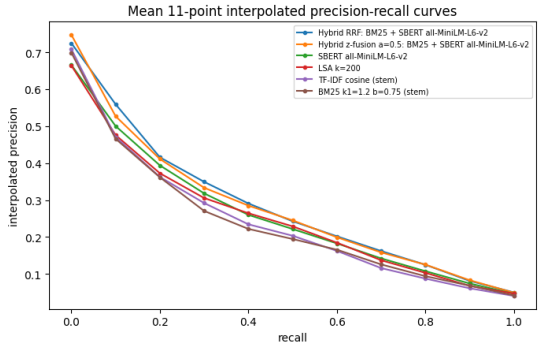


Indexing the title together with the abstract adds 1.4 – 2.5 points over the abstract alone. The variant was chosen on the same queries, so the +0.4-point gap between stemming and lemmatization should be read with caution.

variant	indexed text / model	abstract only	title + abstract
clean	BM25	0.1848	0.2023
	TF-IDF cosine	0.1842	0.2091
stem	BM25	0.2109	0.2250
	TF-IDF cosine	0.2158	0.2291

System	P@5	P@10	F1@10	MAP	nDCG@10
LSA (k = 200)	0.403	0.366	0.170	**0.239**	0.398
Hybrid z-fusion (BM25 + Word2Vec-CISI idf), α = 0.5	**0.434**	0.366	0.170	0.237	**0.411**
Hybrid RRF (BM25 + Word2Vec-CISI idf)	0.421	0.349	0.165	0.231	0.394
TF-IDF cosine (stemmed)	0.408	0.354	0.171	0.229	0.400
BM25 (k1 = 1.2, b = 0.75)	0.429	0.367	0.175	0.225	0.405
TF-IDF cosine, unigrams + bigrams	0.347	0.299	0.147	0.209	0.329
Word2Vec trained on CISI (idf-weighted mean)	0.332	0.316	0.146	0.185	0.340
BM25 with binary query-term frequency	0.347	0.303	0.132	0.179	0.333
GloVe-300 (idf-weighted mean)	0.347	0.291	0.126	0.164	0.323
GloVe-100 (idf-weighted mean)	0.268	0.226	0.094	0.144	0.264





	A	B	mean dAP (A-B)	CI 2.5%	CI 97.5%	Wilcoxon p	A wins / B wins
0	BM25 k1=1.2 b=0.75 (stem)	TF-IDF cosine (stem)	-0.0041	-0.0170	0.0078	0.6749	37 / 39
1	LSA k=200	TF-IDF cosine (stem)	0.0102	-0.0041	0.0278	0.2378	47 / 29
2	SBERT all-MiniLM-L6-v2	BM25 k1=1.2 b=0.75 (stem)	0.0238	0.0001	0.0482	0.0548	45 / 31
3	SBERT all-MiniLM-L6-v2	TF-IDF cosine (stem)	0.0197	-0.0044	0.0437	0.0997	41 / 35
4	Hybrid RRF: BM25 + SBERT all-MiniLM-L6-v2	BM25 k1=1.2 b=0.75 (stem)	0.0469	0.0312	0.0628	0.0000	58 / 18
5	Hybrid RRF: BM25 + SBERT all-MiniLM-L6-v2	SBERT all-MiniLM-L6-v2	0.0231	0.0096	0.0365	0.0005	55 / 21
6	GloVe-300 idf-weighted	GloVe-300 mean	0.0090	-0.0038	0.0223	0.3278	43 / 33

Given that only 76 queries could be evaluated, it is estimated that most of the differences between the top-ranked systems (≤ 1.4 MAP points) fall within the margin of sampling error (confidence intervals, obtained via bootstrapping, overlap significantly); only the differences between models based on word-vector aggregation and the group of sparse, LSA, or hybrid models are clearly significant.

The Wilcoxon p-values have not been corrected for multiple comparisons. The preprocessing variant and the hybrid component were selected using the same queries employed for the reported results (which introduces a slight optimistic bias); hyperparameter tuning was performed via cross-validation on these same queries, except for the rows labeled as "tuned (in-sample)." Unevaluated documents are treated as non-relevant—a common practice, although it penalizes systems for retrieving relevant documents that simply happen not to have been evaluated.

As for the Natural Language queries, we obtained the following results:

```
QUERY: How are computers used to automatically index and classify library documents?
 -> TF-IDF cosine (stem)
  1.  [0.381] doc  662: Automatic Indexing: An Experimental Inquiry
  2.  [0.373] doc  663: Automatic Document Classification Part II. Additional Experiments
  3.  [0.368] doc  564: Automatic Document Classification
  4.  [0.349] doc  565: Computer Evaluation of Indexing and Text Processing
 -> BM25 k1=1.2 b=0.75 (stem)
  1.  [16.711] doc  662: Automatic Indexing: An Experimental Inquiry
  2.  [16.382] doc  564: Automatic Document Classification
  3.  [14.482] doc  663: Automatic Document Classification Part II. Additional Experiments
  4.  [14.177] doc  503: Automatic Classification and Retrieval of Documents by Means of a Bibliographic…
 -> SBERT all-MiniLM-L6-v2
  1.  [0.761] doc  662: Automatic Indexing: An Experimental Inquiry
  2.  [0.724] doc  790: Computer Indexing of Medical Articles - Project Medico
  3.  [0.711] doc  522: Automatic Indexing and Generation of Classification Systems by Algorithm
  4.  [0.701] doc 1280: Searching Natural Language Text by computer
 -> Hybrid RRF: BM25 + SBERT all-MiniLM-L6-v2
  1.  [0.033] doc  662: Automatic Indexing: An Experimental Inquiry
  2.  [0.031] doc  790: Computer Indexing of Medical Articles - Project Medico
  3.  [0.031] doc  522: Automatic Indexing and Generation of Classification Systems by Algorithm
  4.  [0.031] doc 1419: Utility of Automatic Classification Systems for Information Storage and Retrieval
```

```
QUERY: problems of scientific journals growth and how researchers keep up with the literature
 -> TF-IDF cosine (stem)
  1.  [0.273] doc  943: Austerity, Technology, and Resource Sharing: Research Libraries Face the Future
  2.  [0.259] doc  605: Citations, References and the Growth of Scientific Literature: A Model of Dynamic…
  3.  [0.239] doc 1167: Prospects for Primary and Secondary Scientific Publications
  4.  [0.231] doc 1256: The Growth of the Literature of Physics
 -> BM25 k1=1.2 b=0.75 (stem)
  1.  [16.033] doc  943: Austerity, Technology, and Resource Sharing: Research Libraries Face the Future
  2.  [13.849] doc  821: Recent Growth of the Literature of Biochemistry and Changes in Ranking of Periodicals
  3.  [13.177] doc  770: Tests on Abstracts Journals
  4.  [12.636] doc 1167: Prospects for Primary and Secondary Scientific Publications
 -> SBERT all-MiniLM-L6-v2
  1.  [0.667] doc 1167: Prospects for Primary and Secondary Scientific Publications
  2.  [0.665] doc 1168: Analysis of Biological Publication Growth on the Basis of Periodical Sources
  3.  [0.658] doc  605: Citations, References and the Growth of Scientific Literature: A Model of Dynamic…
  4.  [0.650] doc  110: The Gatekeepers of Science: Some Factors Affecting the Selection of Articles of…
 -> Hybrid RRF: BM25 + SBERT all-MiniLM-L6-v2
  1.  [0.032] doc 1167: Prospects for Primary and Secondary Scientific Publications
  2.  [0.031] doc  605: Citations, References and the Growth of Scientific Literature: A Model of Dynamic…
  3.  [0.029] doc  800: The Growth, Utility, and Obsolescence of Scientific Periodical Literature
  4.  [0.028] doc  943: Austerity, Technology, and Resource Sharing: Research Libraries Face the Future
```

```
CISI query 44 with relevance marks (✓ = judged relevant):
 -> BM25 k1=1.2 b=0.75 (stem)
  1. ✓ [39.307] doc  770: Tests on Abstracts Journals
  2.   [39.244] doc  986: MEDLARS: A Summary Review and Evaluation of Three Reports
  3. ✓ [39.168] doc 1460: Modern Integral Information Systems for Chemistry and Chemical Technology
  4. ✓ [37.033] doc  977: A Study of the Usage and Retention of Technical Periodicals
  5.   [36.841] doc  691: Systems Requirements for Primary Information Systems
 -> SBERT all-MiniLM-L6-v2
  1. ✓ [0.700] doc 1167: Prospects for Primary and Secondary Scientific Publications
  2. ✓ [0.634] doc  371: Education and Training for Scientific and Technological Library and Information Work
  3. ✓ [0.620] doc  777: Estimates of the Number of Currently Available Scientific and Technical Periodicals
  4. ✓ [0.610] doc  560: Indivisible Colleges; Diffusion of Knowledge in Scientific Communities
  5. ✓ [0.589] doc  782: Statistics of Scientific and Technical Articles
```

Based on the above, we can conclude that stemming provides more value than the choice between TF-IDF and BM25, as there are no statistically significant

differences between the latter two ($p = 0.67$).

Furthermore, for a small, specialized collection with long queries, it is very difficult to outperform a sparse model that has undergone effective preprocessing.

Finally, LSA and hybrid models yield slight, non-significant improvements in MAP, but they do offer higher precision at the top ranks (P@5, nDCG@10).

Annex 1 – GitHub Repository

https://github.com/Jero2110/NLP_Phase03